



SCHOOL OF APPLIED SCIENCES

A dissertation Report on

**“Multi-Omics Characterization of the Oral-Gut-Brain Axis in
Parkinson’s Disease: Identification of Microbial Translocation
Signatures and Biomarkers”**

Submitted in partial fulfillment of the requirements for the Degree of

MASTER OF SCIENCE

IN

BIOINFORMATICS

Submitted by
PRAVEEN KUMAR S
R24SK017

Under the guidance of
Mr. Rohith Raj Y
Director Of Operations,
Vardhan SK Healthcare Private limited

May 2026

Rukmini Knowledge Park, Kattigenahalli, Yelahanka, Bengaluru-560064

www.reva.edu.in



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DECLARATION

I, **Mr. PRAVEEN KUMAR S** student of M.Sc. Bioinformatics belonging to School of Applied Sciences, REVA University, declare that this internship entitled "**Multi-Omics Characterization of the Oral-Gut-Brain Axis in Parkinson's Disease: Identification of Microbial Translocation Signatures and Biomarkers**" is the result the dissertation work done by me under the supervision of **Mr. Rohith Raj Y** at **VARDHAN SK HEALTHCARE PRIVATE LIMITED**, where the dissertation work has been carried out.

I am submitting this dissertation report for the partial fulfillment of the requirements for the award of the degree of Master of Science in Bioinformatics by the REVA University, Bangalore during the academic year 2025-2026.

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I certify that this dissertation report submitted by **Praveen Kumar S** has been carried out under my guidance and the declaration made by the candidate is true to the best of my knowledge.


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CERTIFICATE

Certified that the dissertation work entitled "**Multi-Omics Characterization of the Oral-Gut-Brain Axis in Parkinson's Disease: Identification of Microbial Translocation Signatures and Biomarkers**" carried out under my guidance by **Mr. Praveen Kumar S (R24SK017)** a bonafide student at REVA University during the year 2025-2026, is being submitted for partial fulfillment of the requirements for the award of the Master of Science in Bioinformatics.

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TO WHOM IT MAY CONCERN

This is to certify that **Mr. Praveen Kumar S** (Registration No. R24SK017), pursuing a Master of Science in Bioinformatics at REVA University, Bengaluru, successfully completed his research project work.

During the project, he worked on the title "Multi-Omics Characterization of the Oral-Gut-Brain Axis in Parkinson's Disease: Identification of Microbial Translocation Signatures and Biomarkers". The study investigated oral-gut microbial dysbiosis associated with Parkinson's disease using shotgun metagenomic sequencing and machine learning approaches.

Mr. Praveen Kumar S demonstrated strong expertise in computational metagenomics, microbiome data analysis, taxonomic profiling, diversity analysis, microbial interaction network construction, functional pathway mapping, virulence factor characterization, and machine learning-based biomarker discovery. His work identified significant microbial signatures associated with Parkinson's disease progression, including oral-to-gut microbial translocation patterns and functional metabolic alterations.

The findings revealed that oral-associated microbial taxa and dysbiosis-linked microbial communities were associated with neuroinflammation, altered metabolic pathways, and disease progression. Machine learning models, particularly XGBoost, demonstrated strong predictive performance for Parkinson's disease classification using microbial biomarkers and Oral Enrichment Score (OES)-based features.

His contribution significantly enhanced the understanding of the oral-gut-brain axis and highlighted the utility of microbiome-derived biomarkers and artificial intelligence-driven approaches in neurodegenerative disease research.

We wish him every success.

Yours Sincerely,



Mr. Rohith Raj Y
Director of Operations,
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PSO3. Learn and utilize scripting languages in genomic data science algorithm development and pipeline design of a wide array of technical research skills.

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2. Students will develop programming skills in languages commonly used in bioinformatics, such as Python, R, and SQL, enabling them to implement algorithms, manipulate datasets, and automate analysis pipelines.
3. Students will integrate heterogeneous biological datasets (e.g., genomics, proteomics, metabolomics) using bioinformatics techniques, and effectively visualize complex data to derive meaningful biological insights.
4. Students will implement and optimize bioinformatics algorithms and statistical methods for tasks such as sequence alignment, gene expression analysis, phylogenetic inference, and protein structure prediction.
5. Students will design and execute a bioinformatics project, applying computational methods to address a specific biological question or problem, demonstrating project planning, execution, and documentation skills.
6. Students will collaborate in teams to complete bioinformatics projects, demonstrating effective communication of methods, results, and implications to diverse audiences, including peers and stakeholders.

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ABBREVIATION OF NOMENCLATURE AND LIST OF SYMBOLS

PD	Parkinson's Disease
PD-MCI	Parkinson's Disease with Mild Cognitive Impairment
PDD	Parkinson's Disease Dementia
AUC	Area Under the Curve
Bracken	Bayesian Re estimation of Abundance with Kraken
DNA	Deoxyribonucleic Acid
GC	Guanine-Cytosine
HC	Healthy Control
LPS	Lipopolysaccharide
MGBA	Microbiota-Gut-Brain Axis
ML	Machine Learning
OES	Oral Enrichment Score
OTU	Operational Taxonomic Unit
PCA	Principal Component Analysis
PCoA	Principal Coordinate Analysis
SVM	Support Vector Machine
VFDB	Virulence Factor Database
XGBoost	Extreme Gradient Boosting
PR-AUC	Precision-Recall Area Under the Curve
QC	Quality Control
RF	Random Forest
RNA	Ribonucleic Acid
ROC	Receiver Operating Characteristic
ROC-AUC	Receiver Operating Characteristic -Area Under Curve
SCFA	Short-Chain Fatty Acid
SMOTE	Synthetic Minority Over-sampling Technique

ABSTRACT

Parkinson's disease (PD) is a progressive neurodegenerative disorder associated with motor dysfunction, neuroinflammation, and microbial dysbiosis linked to the microbiota-gut-brain axis (MGBA). Although gut microbial alterations in PD have been extensively studied, the contribution of the oral microbiome and oral-to-gut microbial translocation mechanisms remains poorly understood. This study aimed to investigate oral-gut microbial interactions associated with Parkinson's disease using an integrated shotgun metagenomics and machine learning framework.

Publicly available paired oral and gut shotgun metagenomic datasets (PRJEB79944) were analysed using a comprehensive bioinformatics pipeline involving quality control, host read removal, taxonomic profiling, diversity analysis, Oral Enrichment Score (OES) calculation, functional pathway mapping, virulence characterization, microbial network analysis, and machine learning-based biomarker discovery. Quality assessment using FastQC and MultiQC demonstrated high-quality sequencing reads with minimal host contamination following preprocessing with fastp, KneadData, and fastq_screen.

Taxonomic profiling using Kraken2 and Bracken identified significant microbial dysbiosis associated with Parkinson's disease progression. Several oral-associated taxa including *Streptomyces albidoflavus*, *Porphyromonas endodontalis*, *Akkermansia* sp. BCRC 18949, and multiple *Streptococcus* species were strongly associated with disease progression. Alpha and beta diversity analyses demonstrated progressive microbial community alterations across clinical groups, while increased OES values in PD-MCI and PDD groups suggested enhanced oral-to-gut microbial translocation.

Functional and virulence analyses revealed enrichment of pathways associated with lipopolysaccharide biosynthesis, amino acid metabolism, oxidative stress, biofilm formation, adhesion, immune modulation, and toxin-associated activities. Machine learning-based classification using XGBoost achieved superior predictive performance with an accuracy of 93%, ROC-AUC of 0.97, and PR-AUC of 0.96.

Overall, this study highlights the mechanistic role of the oral-gut-brain axis in Parkinson's disease and demonstrates the utility of microbiome-derived biomarkers and artificial intelligence-driven approaches for disease prediction and biomarker discovery.

Keywords: Parkinson's disease; Oral-gut-brain axis; Shotgun metagenomics; Oral Enrichment Score (OES); Microbial biomarkers; Machine learning.

INTRODUCTION

1.1 Overview of Parkinson's Disease

Parkinson's disease is a prevalent neurological condition that affects millions of people worldwide. Motor symptoms, including bradykinesia, rigidity, and other motor symptoms are driven by the primary degeneration of dopaminergic neurons, particularly in the substantia nigra region of the brain (Sampson et al., 2016). Additionally, the illness is involved in non-motor symptoms like depression, gastrointestinal problems, sleep disturbances, systemic inflammation, and cognitive deterioration.

Parkinson disease is more common around the world as result of environmental exposure, Ageing population and risk factors on lifestyles. Recent findings suggests that neurodegeneration involves interconnected relationship between environmental, genetic factors and immunological and microbial variables still pathophysiology of Parkinson's disease is still not fully understood (Cryan et al., 2019).

1.2 Microbiota-Gut-Brain Axis in Neurodegeneration

The microbiota-gut-brain axis is a two-way communication system that links gut microbiota, Immune system, central nervous system, enteric nervous system and gastrointestinal tract (Cryan et al., 2019). Multiple mechanisms involvement is observed in this communication system that includes inflammatory mediators, microbial metabolites, neurotransmitter production and vagus nerve stimulation.

For the Parkinson disease progression is caused by microbial alterations in gut that is shown in recent studies and that can influence intestinal permeability, α -synuclein aggregation, neuroinflammation (Sampson et al., 2016).

In Parkinson disease several studies on microbiome shown that upregulation of inflammatory taxa includes *Enterobacteriaceae*, *Akkermansia* and *Bacteroides*, and downregulation of beneficial bacteria involved in short-chain fatty acid production such as *Faecalibacterium* and *Roseburia* (Wallen et al., 2022; Vogt et al., 2017). These alterations in microbes are linked with chronic neuroinflammation, gut barrier integrity impairment and intestinal inflammation.

1.3 Role of Oral Microbiome in Parkinson's Disease

The systemic immunity, microbial pathways and inflammatory signaling are altered by diverse microbes present in the oral cavity. And ongoing research also suggest for the neurodegenerative diseases oral microbes play a very important role by oral to gut translocation pathways (Dominy et al., 2019).

Porphyromonas gingivalis, *Treponema denticola*, *Fusobacterium nucleatum*, and *Streptococcus* species were some of oral linked pathogens that enables host colonization and persistence by their virulence mechanisms. These includes toxin generation, biofilm formation, adhesion factors and immune regulations (Olsen & Singhrao, 2020).

Porphyromonas gingivalis-associated toxic proteases were found in Alzheimer's disease brain tissues by Dominy et al. (2019), indicating a potential link between neurodegeneration and oral pathogens. And also, the gut ecology of microbes was influenced by oral microbes through salivary transmission of oral microbes to gut. However, the oral microbial translocation still unexplored in Parkinson disease.

1.4 Functional and Metabolic Alterations Associated with Microbial Dysbiosis

For maintaining brain and intestinal health microbial metabolites play a very important role. Butyrate, propionate and acetate are some of the examples of short chain fatty acids (SCFA) that control immune regulations, intestinal barrier integrity, anti-inflammatory reactions and neurotransmitter formations (Cryan et al., 2019). So, any dysregulations in metabolic pathways of microbes leads to development of neurodegenerative diseases and long-term inflammations. So, any changes in microbial processes linked to amino acid metabolism, oxidative stress, biofilm formations, lipopolysaccharide production and mucin breakdown have been recorded in Parkinson's disease (Wallen et al., 2022). These modulation in pathways elevate inflammatory signaling and host tissue damage.

The role of virulence linked microbial processes such as toxin generation, immunological regulation, secretion systems, adhesions in host-microbe interactions and disease pathogenesis is also widely acknowledged. However, there is still lack of extensive research on the functional pathways and microbial virulence factors in Parkinson's disease.

1.5 Multi-Omics and Machine Learning in Microbiome Research

Microbial communities can be classified at taxonomic and functional level very precisely by shotgun metagenomic sequencing advancements. Analysis of microbial abundance, metabolic pathways, virulence factors, and microbial interactions is made more easier by computational tools like Kraken2, Bracken, MetaPhlAn, Meta Cyc and HUMAnN (Beghini et al., 2021; Franzosa et al., 2018). For the prediction of diseases and biomarker discovery in microbiome research machine learning techniques are Assumed with greater significance. Random Forest and Extreme Gradient Boosting (XGBoost) algorithms shown strong predictive performance in disease identification of disease associated microbes (Muller et al., 2022).

In neurodegenerative diseases for non-invasive biomarker discovery and precision diagnostics integration of AI with microbiome analytics provides broader opportunities. Despite these developments, most of the research aims in gut microbiome alterations and leaving oral-gut microbial translocations and their interactions largely unexplored. And very few integrative analysis that connects pathways, metabolites, microbial biomarkers, disease development and virulence factors.

REVIEW OF LITERATURE

2. REVIEW OF LITERATURE

Parkinson disease is a neurodegenerative disorder is associated with cognitive decline, movement dysfunction and systematic inflammation. The role of microbial dysbiosis and microbiota-gut-brain-axis in neurodegenerative disorders is confirmed by advancement of recent microbiome research.

Studies increasingly demonstrate both oral and gut microbial communities play role in metabolic changes, neuroinflammation, immune modulations, disease progression.

This chapter describes earlier research on oral microbial translocation, virulence-linked processes, functional microbial pathways, gut microbiome dysbiosis, and machine learning-based biomarker discovery in Parkinson's disease.

2.1 Gut Microbiome Dysbiosis in Parkinson's Disease

Several studies reported significant gut microbial alterations in Parkinson disease, supporting their involvement of microbiota-gut-brain axis in progression of disease. (Sampson et al., 2016) highlighted that changes in the gut microbiota modulates the motor dysfunction and neuroinflammation in mouse model with Parkinson disease suggesting there is a direct link of microbial community with severity of disease.

(Wallen et al., 2022) reported that beneficial SCFA producing bacteria such as *Faecalibacterium* and *Roseburia* were downregulated and inflammatory taxa like *Enterobacteriaceae*, *Bacteroides* and *Akkermansia* were shown upregulated. Functional analysis confirmed higher levels of pathways linked with oxidative stress, inflammatory metabolism and lipopolysaccharide biosynthesis.

In Parkinson disease patient's beneficial microbes like *Faecalibacterium* species and *Prevotella copri* species abundance were reduced and opportunistic pathogens like *Parabacteroides merdae*, *Bilophila wadsworthia*, and *Alistipes onderdonkii* species were high in numbers. This study suggested that dysregulation of microbial metabolite that leads to neurodegenerative progression in Parkinson disease.

2.2 Microbiota-Gut-Brain Axis and Neuroinflammation

The microbiome-gut-brain axis is a network of bidirectional communication which involves central nervous system, immune signaling pathways, microbial metabolites and gastro intestinal tract. Microbial metabolites play crucial role in neuro immune regulation and neuronal signaling mechanisms. The dysbiosis in gut microbiota in Parkinson disease is linked to systemic inflammation, α -synuclein aggregation and intestinal barrier dysfunctions (Cryan et al., 2019).

Chronic neuroinflammation is elevated by LPS secreted by gram-negative bacteria and gut barrier integrity was dysregulated by lower abundance of short-chain fatty acid producing bacteria. In Parkinson's disease the changes in amino acid metabolism, neurotransmitter metabolism and inflammatory pathways were determined which providing the role of microbial metabolite functional impairments in neurodegeneration (Jo et al., 2022).

2.3 Oral Microbiome and Oral-Gut Axis in Neurodegeneration

The oral microbes have ability to produce toxins, form persistence biofilms and modulate immune responses so play a significant role in neurodegeneration and systematic inflammation. The disease like Periodontitis and chronic oral infections may disrupt the epithelial barriers and enables the translocation of microbes from oral to gut and brain and mediating neurodegeneration (Olsen and Singhrao, 2020).

Studies highlights oral microbial dysbiosis in Parkinson's disease with increased abundance of *Capnocytophaga*, *Megasphaera* and *Lactobacillus* along with decrease in *Lautropia* and *Abiotrophia* species (Jo et al., 2022). And suggests that there is a fictional oral-gut microbial associations. The oral-gut-brain axis findings paved the way for oral microbial changes can alter gut microbial ecology and promote inflammation and contribute to neurodegeneration (Jo et al., 2022).

2.4 Oral Pathogens, Virulence Factors, and Neurodegeneration

Stephen N. Dominy et al. (2019) indicated *Porphyromonas gingivalis* an oral pathogen is linked with chronic periodontitis and leads to neurodegeneration in Alzheimer's disease. The study discovered gingipain proteases and *P. gingivalis* DNA in affected patient's brain and cerebrospinal fluid with gingipain levels strongly connected to tau pathology.

Further neuronal damage, brain colonization, amyloid- β production, neuroinflammation was promoted by oral pathogens and were confirmed by experimental models. The neurodegenerative effects were reduced by gingipain inhibitors which uncovered the clinical relevance of targeting microbe's virulence factors. All these indicates mechanistic link of oral pathogens with neurodegenerative disease development facilitated by microbial dissemination, blood brain dysfunction and inflammation.

2.5 Virulence Signatures and Cognitive Decline in Parkinson's Disease

Clasen et al. (2025) demonstrated that oral and gut microbiome patterns were linked with cognitive impairments in Parkinson's disease by using paired shotgun metagenomic sequencing datasets. The research incorporated patients with Parkinson disease with mild cognitive impairments (PD-MCI), Parkinson disease with dementia and Healthy controls.

During cognitive decline there is a significant alteration was observed in both oral and gut microbes and also butyrate producing bacteria were decreased in umbers such as *Roseburia* species, *Faecalibacterium* and *prausnitzii*. And in Parkinson patients with dementia more prevalent microbial species reported are *Desulfovibrio*, *Hungatella*, *Acidaminococcus* and *Olsenella*.

Study also reported that oral pathogens and virulence associated pathways were linked with immune dysregulation, oxidative stress, immune dysregulation and blood brain barrier inflammations. Functional analyses revealed dominance of inflammatory metabolic pathways, antimicrobial resistance mechanisms, amino acid transport system and biofilm formation mechanisms.

Collectively Clasen et al. (2025) study emphasized the significance of oral-gut microbial translocation and virulence linked microbial patterns in advancement of Parkinson disease and cognitive decline.

2.6 Functional Pathways and Metabolic Dysregulation

Franzosa et al. (2018) highlighted that by comparing to microbial taxonomic composition alone the species level functional profiling give deeper mechanistic view and studies in Parkinson disease reported that overrepresentation of pathways linked with oxidative stress, Inflammatory metabolism, mucin degradation, amino acid metabolism and lipopolysaccharide formation.

For neuroinflammation and disease development the changes in mechanisms such as arginine biosynthesis, glutamate metabolism and reduced butyrate production were majorly contributed Clasen et al. (2025).

Over all these study reveals the importance of combining metabolic and functional analysis for characterization of host-microbes association in neurodegenerative disorders.

2.7 Machine Learning for Microbiome-Based Biomarker Discovery

Machine learning approaches have gained increasing importance in microbiome research for disease prediction, classification, and biomarker discovery. Algorithms such as Random Forest, Support Vector Machines (SVM), and XGBoost have shown strong predictive capability in identifying disease-associated microbial signatures.

Clasen et al. (2025) integrated oral and gut microbiome profiles with clinical metadata using machine learning models to predict Parkinson's disease-associated cognitive impairment. The study identified microbial biomarkers including *Faecalibacterium prausnitzii*, *Roseburia faecis*, and *Treponema denticola* as significant predictive features linked to cognitive decline.

Overall, machine learning-based microbiome analyses enhance disease classification accuracy and support the identification of biologically relevant microbial biomarkers for precision medicine applications.

OBJECTIVE

- 1) To investigate oral and gut microbial dysbiosis associated with Parkinson's disease using shotgun metagenomic analysis.
- 2) To identify oral-to-gut microbial translocation signatures, functional pathways, and virulence-associated mechanisms involved in the oral-gut-brain axis.
- 3) To develop and evaluate machine learning models for Parkinson's disease classification using Oral Enrichment Score (OES) for early disease prediction and identification of microbial biomarkers associated with Parkinson's disease.

MATERIALS REQUIRED

4. Materials Required:

4.1 Biological Datasets

- Publicly available paired oral and gut shotgun metagenomic datasets
- Parkinson's disease and healthy control microbiome samples
- Dataset Project ID PRJEB79944 from NCBI Sequence Read Archive (SRA)
- Human Oral Microbiome Database (HOMD)

4.2 Computational Resources

- Linux-based workstation/server
- High-performance computing environment
- Sufficient storage for shotgun metagenomic datasets
- Internet access for database retrieval and software installation

4.3 Quality Control and Preprocessing Tools

- FastQC
- MultiQC
- fastp
- KneadData
- fastq_screen

4.4 Taxonomic Profiling Tools

- Kraken2
- Bracken
- Krona

4.5 Feature Engineering and Statistical Tools

- Python programming environment
- R programming environment
- Oral Enrichment Score (OES) calculation pipeline

4.6 Functional and Virulence Analysis Tools

- Virulence Factor Database (VFDB)
- MetaCyc pathway database
- Functional annotation and pathway mapping tools

4.7 Network Analysis Tools

- PyVis network visualization tool

4.8 Machine Learning and Biomarker Discovery Tools

- XGBoost algorithm
- Random Forest algorithm
- SMOTE for class balancing
- Biomarker analysis pipeline
- Scikit-learn library

4.9 Visualization and Documentation Tools

- Microsoft Excel
- Matplotlib and Seaborn visualization libraries

METHODOLOGY

5. METHODOLOGY

5.1 Study Design

The current study was designed to investigate the oral-gut-brain axis in Parkinson's disease using an integrated multi-omics and machine learning approaches. For that Publicly available paired oral and gut shotgun metagenomic datasets of total 241 samples of Parkinson's disease patients with healthy controls were analysed to identify microbial alterations, oral-to-gut microbial translocation patterns, functional pathways, virulence-associated mechanisms, and predictive microbial biomarkers linked with disease progression.

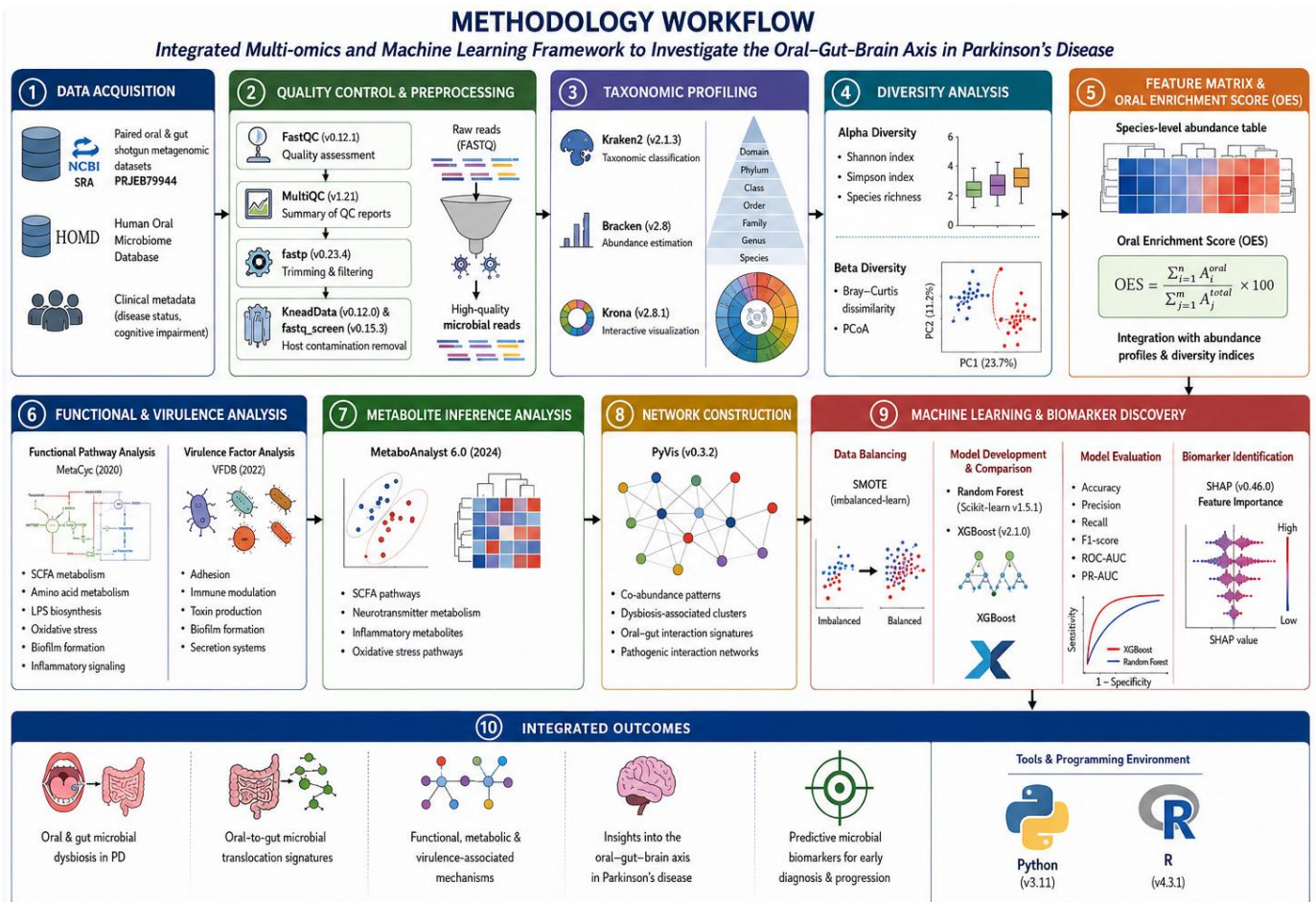


Figure 1. Integrated methodology workflow used for oral-gut microbiome analysis in Parkinson's disease.

5.2 Data Acquisition

Publicly available datasets with Project ID PRJEB79944 containing paired oral and gut shotgun WGS datasets were obtained from NCBI SRA database. Associated clinical metadata including disease status and cognitive impairment information were also retrieved.

For identifying oral-associated microbial species and evaluation of oral microbial enrichment patterns. The reference database Human Oral Microbiome Database was used.

5.3 Quality Control and Preprocessing

Raw sequencing reads obtained from shotgun metagenomic datasets were subjected to quality assessment and preprocessing prior to downstream analysis.

5.3.1 Quality Assessment

FastQC v0.12.1 (Andrews, 2010) was used for the evaluation of the sequence quality and the metrics includes:

- Per base sequence quality
- GC content distribution
- Sequence duplication levels
- Adapter contamination
- Overrepresented sequences

MultiQC v1.21 (Ewels et al., 2016) was used for summarizing and visualization of combined sample reports.

5.3.2 Read Trimming and Filtering

Low-quality bases, sequencing adapters, and poor-quality reads were removed using fastp v0.23.4 (Chen et al., 2018). Reads with low Phred quality scores and short read lengths were filtered to improve downstream analytical accuracy.

5.3.3 Host Contamination Removal

Human host contamination was removed using KneadData v0.12.0 and fastq_screen v0.15.3 by aligning sequencing reads against the human reference genome. Only non-host microbial reads were retained for microbiome analysis.

5.4 Taxonomic Profiling

Kraken2 v2.1.3 (Wood et al., 2019) tool was used for taxonomic classification its basically k-mer based classification of metagenomic sequence classification. And Kraken2 identifies bacteria at various taxonomic levels including phylum, genus and species by classifying sequences against the standard reference databases. Bracken v2.8 (Lu et al., 2017) was subsequently used for abundance estimation and refinement of Kraken2 classifications to obtain accurate species-level abundance profiles.

Krona v2.8.1 (Ondov et al., 2011) plots were generated for interactive visualization of microbial taxonomic composition across oral and gut microbiome samples.

5.5 Diversity Analysis

Microbial diversity analyses were performed to evaluate community composition and dysbiosis patterns associated with Parkinson's disease.

5.5.1 Alpha Diversity Analysis

Alpha diversity metrics including:

- Shannon diversity index
- Simpson diversity index
- Species richness

were calculated to evaluate within sample diversity of microbes.

5.5.2 Beta Diversity Analysis

To compare microbial diversity between Parkinson disease and Healthy control clinical groups Beta diversity analysis was performed using distance-based methods.

And to visualize microbes across clinical groups Principal Coordinate Analysis and Bray-Curtis dissimilarity metrics were used.

5.6 Feature Matrix Construction and Oral Enrichment Score (OES)

For the construction of feature matrix from microbial abundance table the taxonomic profiles were processed using Python v3.11 and R v4.3.1.

To quantify oral linked microbial signatures in gut samples a novel oral enrichment score (OES) was developed. By using HOMD database oral associated taxa were identified and were used for calculating enrichment scores to evaluate oral-to-gut translocation dynamics of microbes.

The OES was integrated with microbial abundance profiles and diversity indices to improve disease classification and biomarker identification.

5.7 Functional and Virulence Analysis

5.7.1 Functional Pathway Analysis

Functional metabolic pathway analysis was performed using MetaCyc database mapping (Caspi et al., 2020) to identify microbial pathways associated with Parkinson's disease progression.

Pathways related to:

- Short-chain fatty acid metabolism
- Amino acid metabolisms
- Lipopolysaccharide production
- Oxidative stress
- Biofilm formation
- Inflammatory signaling

were investigated.

5.7.2 Virulence factor Analysis

Virulence related microbial genes were examined using the Virulence Factor Database (Liu et al., 2022).

Virulence mechanisms including:

- Adhesion
- Immune modulation
- Toxin production
- Biofilm formation
- Secretion systems

were evaluated to understand microbial pathogenicity associated with oral-gut translocation and neurodegeneration.

5.8 Network Construction

Microbial interaction networks were constructed using PyVis v0.3.2 for visualization of microbial relationships associated with Parkinson's disease.

Correlation-based microbial interaction analysis was performed to identify:

- Co-abundance patterns
- Dysbiosis-associated microbial clusters
- Oral–gut microbial interaction signatures
- Potential pathogenic interaction networks

associated with disease progression.

5.9 Machine Learning and Biomarker Discovery

Machine learning models were developed to classify Parkinson's disease samples and identify predictive microbial biomarkers.

5.9.1 Data Balancing

Synthetic Minority Oversampling Technique (SMOTE) implemented in the imbalanced-learn Python library was applied to address class imbalance within the dataset and improve model performance.

5.9.2 Model Development

Random Forest and Extreme Gradient Boosting machine learning Algorithms were used for comparative model evaluation for the classification of Parkinson disease. Random Forest model was developed using Scikit-learn v1.5.1 (Pedregosa et al., 2011), And XGBoost v2.1.0 (Chen & Guestrin, 2016) was utilized for improved gradient boosting based feature optimization and classification.

The models were trained using:

- Microbial abundance profiles
- Diversity indices
- Oral Enrichment Score (OES)
- Functional pathway features
- Virulence-associated microbial signatures

5.9.3 Comparative Model Evaluation

The predictive performance of Random Forest and XGBoost models was compared using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- PR-AUC

To evaluate model robustness and generalization cross-validation analysis was demonstrated.

5.9.4 Biomarker Identification

Feature importance analysis was performed to identify microbial taxa, pathways, and OES-associated signatures strongly associated with Parkinson's disease progression and oral-gut microbial translocation.

Important biomarkers identified by the machine learning models were visualized using SHAP v0.46.0 (Lundberg & Lee, 2017) analysis and feature ranking approaches.

5.10 Statistical Analysis

Statistical analyses were performed using Python v3.11 and R v4.3.1 programming environments. Differential abundance analysis and diversity comparisons between clinical groups were conducted using appropriate statistical tests.

P-values less than 0.05 were considered statistically significant throughout the study.

RESULTS AND DISCUSSION

6.1 Quality Control and Preprocessing Analysis

6.1.1 Sequence Count Analysis

Sequence count analysis generated using Fastqc tool showed significant variation in the sequencing depth and duplicate read distribution across samples. High number of unique reads were present in majority of samples denoting high sequencing coverage. And some samples showed relatively higher duplicate read counts, and in majority of datasets unique reads remained dominant. Suggesting overall quality and reliability of sequencing data.

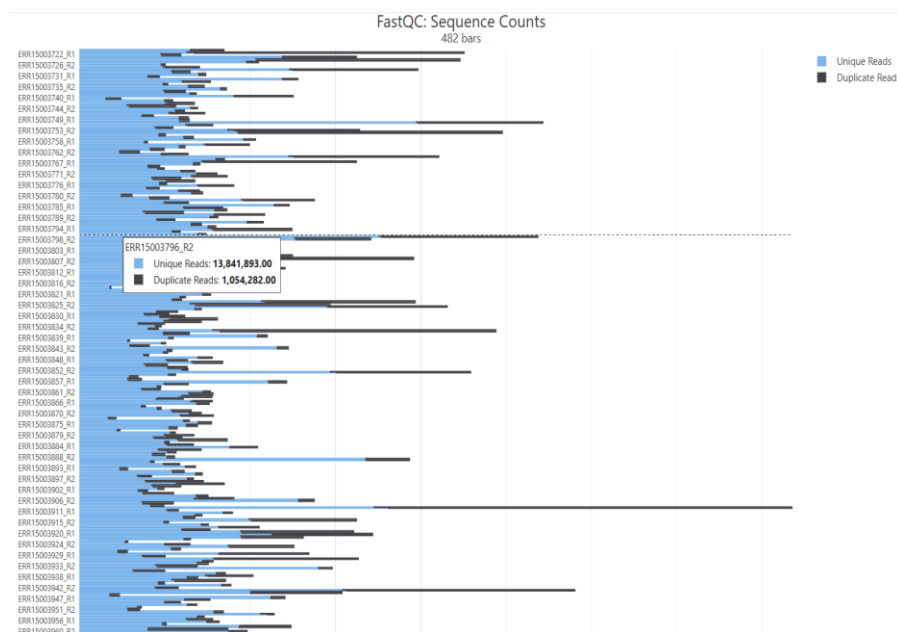


Figure 2. FastQC sequence count analysis of shotgun metagenomic samples.

For accurate microbial profiling and metagenomic downstream analysis sufficient read depth is important. In the present study domination of unique reads were observed that suggests efficient sequencing performance across samples. In duplicated reads some variations were observed that may be linked to amplification bias and due to differences in microbial abundance. Overall, the distribution of sequence counts confirmed that samples are suitable for downstream analysis.

6.1.2 Per Sequence GC Content Analysis

Per sequence GC content derived from FASTQC tool showed normal GC distribution across samples. The most of the samples showed peaks between 40-60%, suggesting stable microbial representation and consistent sequencing composition across the datasets. GC content quality assessment is passed by most of samples and some samples denoted those moderate deviations from expected distribution patterns.

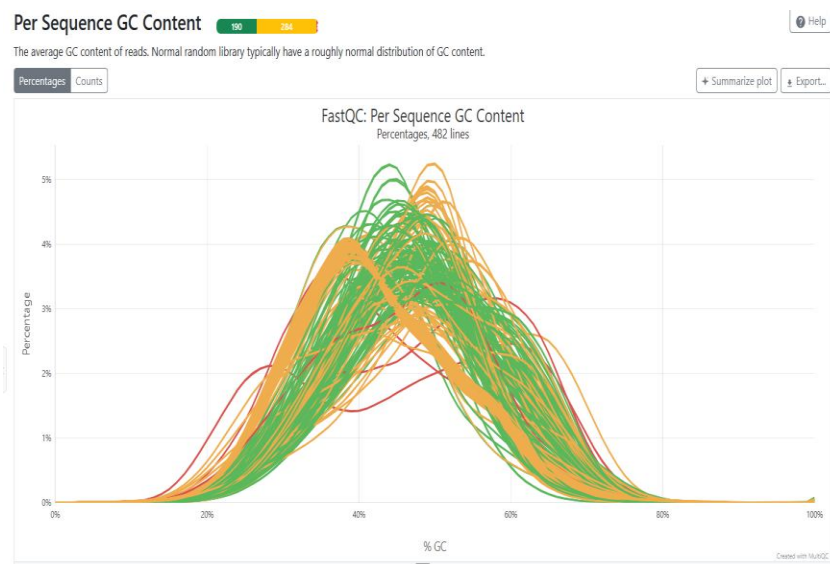


Figure 3. FastQC per-sequence GC content analysis across metagenomic samples.

In metagenomic studies to examine the sequencing bias, contamination and consistency of datasets GC content analysis is key quality control parameter. In current study most of samples showed normal GC distribution suggesting good quality of sequence and less technical bias. The GC range of 40-60% indicates reliable bacterial communities in the sample. Overall GC content analysis confirmed that datasets is suitable for further downstream analysis.

6.1.3 MultiQC Summary Statistics

MultiQC summary statistics given overview of sequence quality across all samples. The significance variation in Duplication levels, GC content and sequencing depth is determined. Majority of samples shown GC content ranging in 40-50% which suggesting stable microbial compositions across study. some samples containing sequencing depth more than 40 million reads indicated sequencing depth variation across samples. And in most of samples the levels of duplication are low to moderately ranged. overall, the MultiQC summary statistics suggested acceptable sequence quality and excellent read coverage for further downstream analysis.

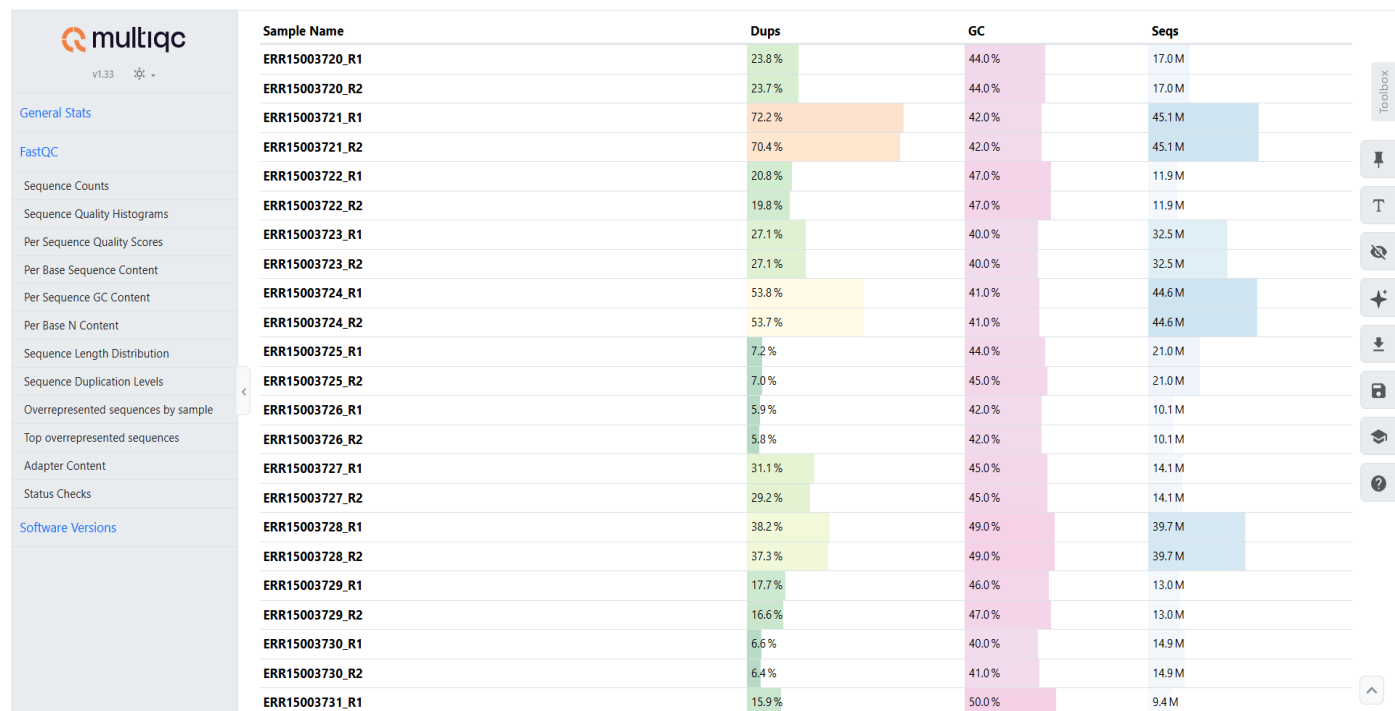


Figure 4. MultiQC summary statistics showing duplication levels, GC content, and sequencing depth across metagenomic samples.

MultiQC gives a combined view of sequencing quality metrics and is use full for detecting inconsistencies in large metagenomic datasets. In the current study GC content distribution is in expected range implies reliability of sequencing data. In shotgun metagenomic studies variation in sequencing depth is more common that shows imbalance in microbial biomass and sequencing efficiency. Although some samples show higher level of duplications, the major samples maintained reliable sequencing coverage and acceptable quality metrics. High sequencing depth maximizes the sensitivity of microbial detection and increases the accuracy of downstream analysis

6.1.4 Host Contamination Screening Analysis

Host contamination screening analysis conducted using FASTQ SCREEN and summarized by MULTIQC. To detect the presence of human contamination in metagenomic datasets. The analysis revealed that majority of reads were categorized as “No Hits” indicating that host-derived contaminations were removed successfully during preprocessing. Across the samples alignment to the human genome is very minimal so which confirms the efficient filtering of human reads and retention of non-host microbial reads for downstream analysis.



Figure 5. Host contamination screening analysis of cleaned metagenomic samples following preprocessing and host read removal.

In shotgun metagenomic studies host decontamination is the important preprocessing step for accurate microbial profiling and reduce host derived sequence bias. In this present study fastq_screen analysis showed effective Removal of human genomic contaminations. Minimal host alignments observed in some samples may be due to less sequence similarity these are observed in complex metagenomic datasets.

Taken together preprocessing efficiently retained high quality microbial reads.

6.2 Taxonomic Profiling Using Kraken2 and Bracken

For all the 241 metagenomic samples taxonomic profiling was performed using Kraken2 v2.1.3 and followed by abundance refinement using bracken v2.8. The diverse bacterial taxa were identified across oral and gut samples at multiple taxonomic levels. most of the reads were categorized under domain bacteria with dominance of phylum *Bacteroidota* and class *Bacteroidia*. Highly abundant taxa identified as species level are *Phocaeicola vulgatus*, *Phocaeicola dorei*, *Bacteroides ovatus*, and *Bacteroides uniformis*. Normalized species abundance was generated using bracken from kraken2 classification for downstream diversity, functional and biomarker analysis.

0.31	141490	141490	U	0	unclassified
99.69	44907804	5603	R	1	root
99.67	44900777	122	R1	131567	cellular organisms
99.67	44899897	91619	R2	2	Bacteria
98.21	44241199	388938	K	3379134	Pseudomonadati
92.80	41806766	0	K1	1783270	FCB group
92.80	41806760	6	K2	68336	Bacteroidota/chlorobiota group
92.80	41806746	38293	P	976	Bacteroidota
92.71	41767311	63	C	200643	Bacteroidia
92.71	41767200	708517	O	171549	Bacteroidales
90.87	40934196	1056881	F	815	Bacteroidaceae
67.25	30294610	2460535	G	909656	Phocaeicola
60.57	27286150	27285662	S	821	Phocaeicola vulgatus
0.00	488	488	S1	435590	Phocaeicola vulgatus ATCC 8482
1.19	534491	527887	S	357276	Phocaeicola dorei
0.01	6604	6604	S1	997877	Phocaeicola dorei CL03T12C01
0.03	11374	11374	S	387090	Phocaeicola coprophilus
0.00	2050	0	S	376805	Phocaeicola salanitronis
0.00	2050	2050	S1	667015	Phocaeicola salanitronis DSM 18170
0.00	10	0	S	310298	Phocaeicola coprocola
0.00	10	10	S1	470145	Phocaeicola coprocola DSM 17136
21.27	9582705	4602732	G	816	Bacteroides
7.87	3545299	3543215	S	28116	Bacteroides ovatus
0.00	2084	2084	S1	1379690	Bacteroides ovatus V975
0.80	361834	352595	S	820	Bacteroides uniformis
0.02	9239	9239	S1	997890	Bacteroides uniformis CL03T12C37

Figure 6. Kraken2 taxonomic classification output showing hierarchical microbial composition across metagenomic samples.

name	taxonomy_id	taxonomy_lvl	kraken_assigned_reads	added reads	new_est reads	fraction_total_reads
Phocaeicola vulgatus	821	S	27286150	4279624	31565774	0.70300
Phocaeicola dorei	357276	S	534491	72810	606501	0.01351
Phocaeicola coprophilus	387090	S	11374	572	11946	0.00027
Phocaeicola salanitronis	376805	S	2050	19	2069	0.00005
Phocaeicola coprocola	310298	S	10	48	58	0.00000
Bacteroides ovatus	28116	S	3545299	4068199	7613498	0.16956
Bacteroides uniformis	820	S	361834	124184	486018	0.01082
Bacteroides caccae	47678	S	248530	8972	257502	0.00573
Bacteroides xylanisolvens	371601	S	209977	383898	593875	0.01323
Bacteroides fragilis	817	S	197474	86258	283732	0.00632
Bacteroides thetaiotaomicron	818	S	172626	12833	185459	0.00413
Bacteroides sp. CG01	3096000	S	23242	2218	25460	0.00057
Bacteroides sp. KFT8	2025659	S	20458	15108	35566	0.00079
Bacteroides sp. M10	2763022	S	18908	8094	27002	0.00060
Bacteroides sp. D2	556259	S	14221	12461	26682	0.00059
Bacteroides sp. CACC 737	2755405	S	8059	17265	25324	0.00056
Bacteroides sp. DH3716P	2847299	S	7104	36699	43803	0.00098
Bacteroides sp. AIC1	2528203	S	4846	13681	18527	0.00041
Bacteroides sp. HF-162	2785531	S	1212	4906	6118	0.00014
Bacteroides sp. BFG-257	2972761	S	1021	33005	34026	0.00076
Bacteroides sp. KG123	3369153	S	188	12820	13008	0.00029
Bacteroides sp. ZJ-18	2709390	S	35	157	192	0.00000
Bacteroides sp. PHL 2737	2162637	S	13	1594	1607	0.00004
Bacteroides sp. KG68	3307824	S	12	1529	1541	0.00003
Bacteroides caecimuris	1796613	S	19254	2362	21616	0.00048

Figure 7. Bracken species-level abundance estimation and refined microbial quantification profiles.

For understanding microbial community diversity and disease linked microbial changes taxonomic profiling plays significant role. In present study Kraken2 and Bracken analysis demonstrated dominance of *Bacteroidota* associated taxa, which is widely observed in human gut ecosystems. *Phocaeicola vulgatus* and *Bacteroides uniformis* species were higher in levels denoting likely involvement of host-microbe interactions linked with Parkinson disease. The diverse bacterial taxa across oral and gut microbiomes indicates complexity of ecosystem of microbes involved in oral-gut-brain axis.

6.2.1 Krona-Based Taxonomic Visualization

Krona based taxonomic visualization highlighted that most of the reads classified were bacterial domain, dominated with phylum *Bacteroidota* and class *Bacteroidia*. And *Phocaeicola vulgatus*, *Bacteroides ovatus*, *Phocaeicola dorei*, and *Bacteroides uniformis* species were higher in species levels. Across samples viral and unclassified sequences were minimal in composition.

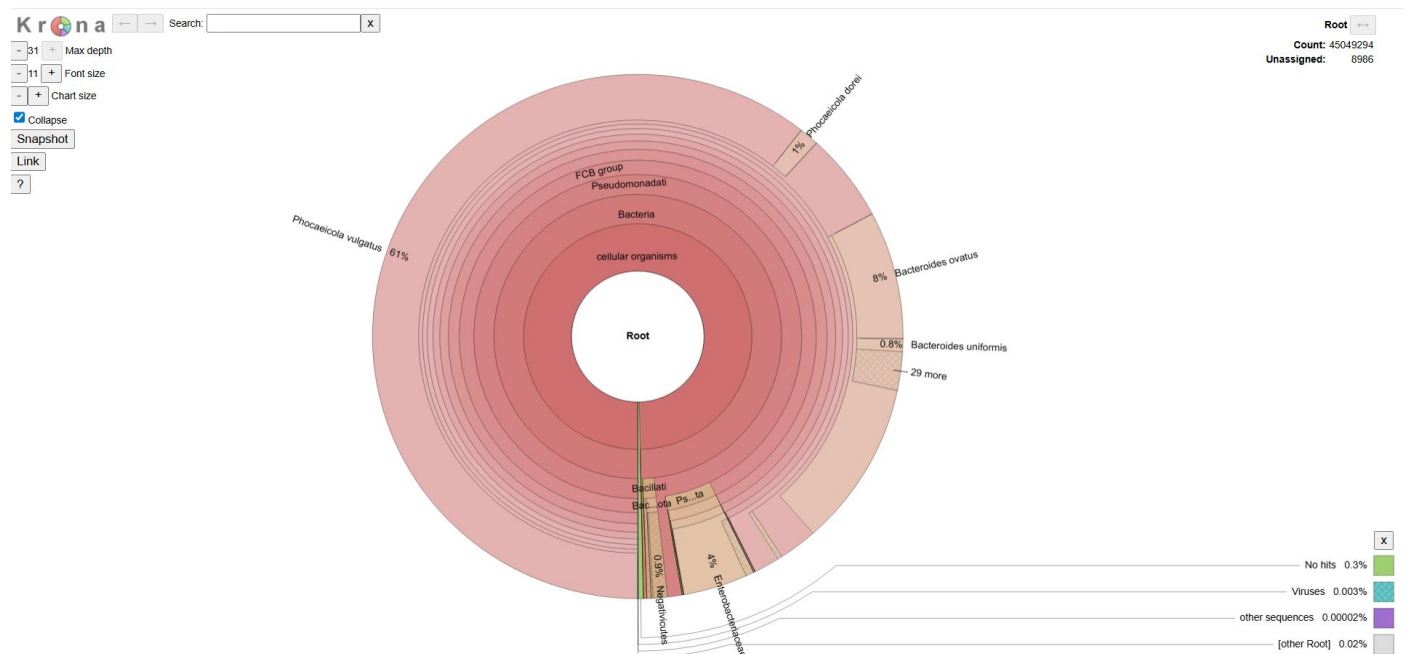


Figure 8. Krona plot illustrating hierarchical taxonomic distribution and relative abundance of dominant microbial taxa

Krona visualization gives an interactive overview of microbial composition at various taxonomic levels. The dominance of *Bacteroidota* linked taxa suggests important role of host metabolic interactions and gut microbial ecology. Highly dominant species such as *Phocaeicola vulgatus* and *Bacteroides ovatus* have been linked with inflammation within gut microbiome. Overall microbial diversity implies dominance of bacterial community and their role in dysbiosis with Parkinson's disease.

6.3 Species Matrix and Relative Abundance Analysis

[illegible]

Figure 9. Relative abundance matrix of gut microbiome species across metagenomic samples.

	0.0008417287864694553	0.0010362133278472858	0.0	0.0005504782279605407	0.002210158581424483	4.504525640604545e-05
	5.944172333444291e-05	0.0005506820925512358	0.0031056559637207457	0.011434577766118482	0.002578250688096643	
	0.003823610676195913	5.477634513964467e-05	0.000456622332613014	0.00023616697368373247	0.00037784270672676693	0.0
Abyssalbus ytuae	0.0010115172759138707	0.001899546842979283	0.0	0.0017929205206206151	8.832058957906482e-05	0.0006935620499788225
0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	1.859160494839115e-06	0.0	0.0	1.7713317048890527e-06
2.3373716687998104e-06	0.0	0.0	0.0	0.0	0.0	0.0
1.196063905911959e-06	0.0	4.0691090075461625e-06	0.0	0.0	0.0	1.6257309547650812e-06
0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0
Abyssiobius alkaniclasticus	0.0	0.0	0.0	5.256644727475825e-06	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0
3.8969297507559155e-06	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0
2.5894330048021034e-06	0.0	2.3247065586312126e-06	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0
Abyssiococcus albus	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	7.203415599882147e-06
0.0	0.0	0.0	0.0	0.0	0.0	9.3494866751199241e-07
0.0	0.0	0.0	0.0	0.0	0.0	6.529070362108161e-05
0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0

Figure 10. Relative abundance matrix of oral microbiome species across metagenomic samples.

Species abundance matrices were constructed using species level taxonomic profiles for oral and gut microbiome datasets from kraken2 and Bracken results. The resulting matrices represented distribution of microbial species and count profiles of all the samples revealing substantial differences in microbial diversity across the datasets. *Acetobacter* species were dominated in gut microbiome samples and *Pasteurella* species were enriched in oral microbiome samples. Across clinical groups the normalized relative abundance compositions provide a systematic comparison of microbial diversity and abundance distribution.

6.4 Oral vs Gut Top Species Mean Relative Abundance Analysis

Comparative assessment of mean relative abundance revealed clear microbial compositional variation between gut and oral samples. *Segatella copri*, *Phocaeicola vulgatus*, *Bacteroides uniformis*, *Bacteroides fragilis*, and *Alistipes* species were showed dominant in gut microbiome datasets. Oral associated taxa such as *Rothia mucilaginosa*, *Haemophilus parainfluenzae*, *Veillonella atypica*, *Streptococcus mitis*, and *Neisseria subflava* were enriched in oral microbiome samples. These indicates distribution of microbes between oral and gut ecosystem.

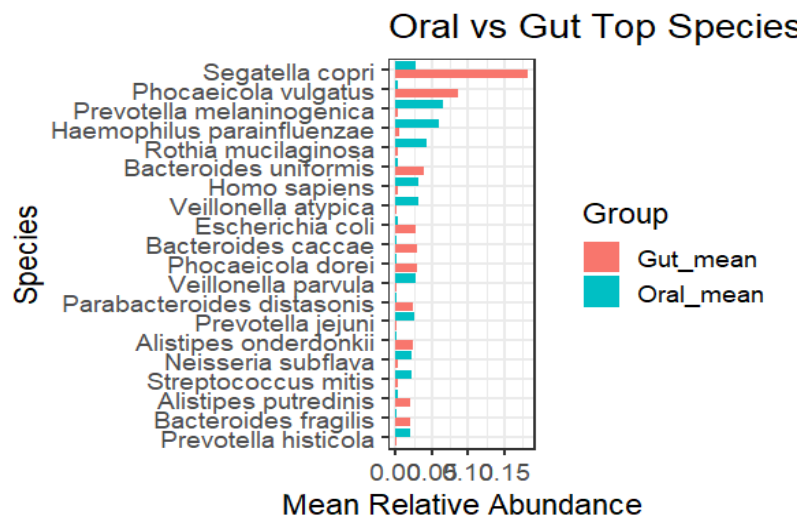


Figure 11. Comparison of top microbial species mean relative abundance between oral and gut microbiome samples.

From the comparative relative abundance there is a potential role of oral-gut microbial interactions linked to Parkinson's disease due to the enrichment of gut microbes linked to intestinal metabolism and host microbe interactions. On the other hand, dominated oral taxa were linked with oral microbial colonization. The presence of oral origin microbes within disease link microbial signatures promotes the possibility of oral to gut microbial translocation and dysbiosis associated with Parkinson disease progression.

6.5 Oral Enrichment Score (OES) Analysis Across Clinical Groups

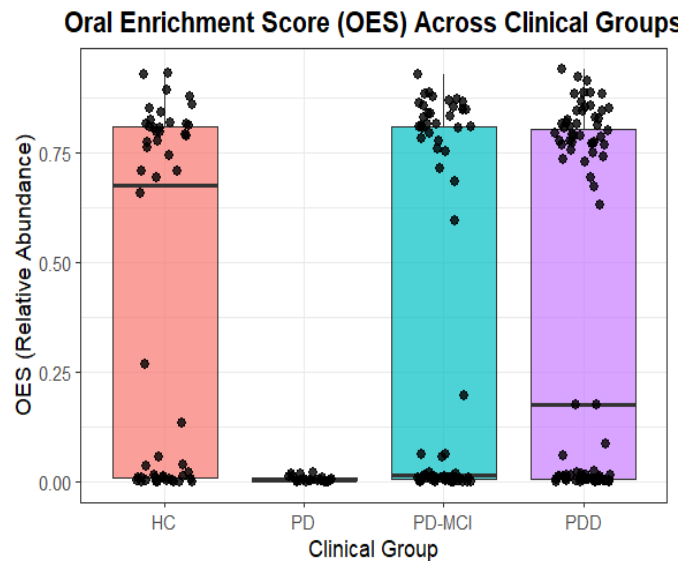


Figure 12. Oral Enrichment Score (OES) distribution across Parkinson's disease clinical groups.

Oral Enrichment Score (OES) analysis highlighted significant alteration in oral microbes in different clinical groups. Healthy control samples showed stable OES distribution however Parkinson disease samples indicated lower OES values. And increased OES distribution was obtained in PD-MCI and PDD groups suggesting variation in enrichment of microbial taxa during disease progression and cognitive decline.

$$\text{OES} = \frac{\sum \text{Relative abundance of oral-associated species}}{\sum \text{Total microbial relative abundance}}$$

The oral Enrichment Score (OES) constructed to evaluate relative abundance of oral associated microbes within gut microbiome. The less OES values in PD samples implies during early stages of disease progression the oral-gut microbial balance is disrupted. But in PD-MCI and PDD groups OES score is high in number suggesting more microbial translocation from oral to gut leading to the cognitive decline and increasing severity of disease by microbial dysbiosis. Variation in OES in different clinical groups suggesting the

practical significance as quantitative microbial biomarker for disease progression and microbial movement patterns in Parkinson disease.

6.6 Microbial Diversity Analysis

Alpha diversity analysis using Shannon diversity index exhibited there is a variation in diversity of microbes across clinical groups. On compared to Healthy controls the Shannon diversity values are lower in Parkinson disease samples. Alternatively, PD-MCI and PDD groups revealed high level of diversity implying progressive changes in microbial diversity during disease development.

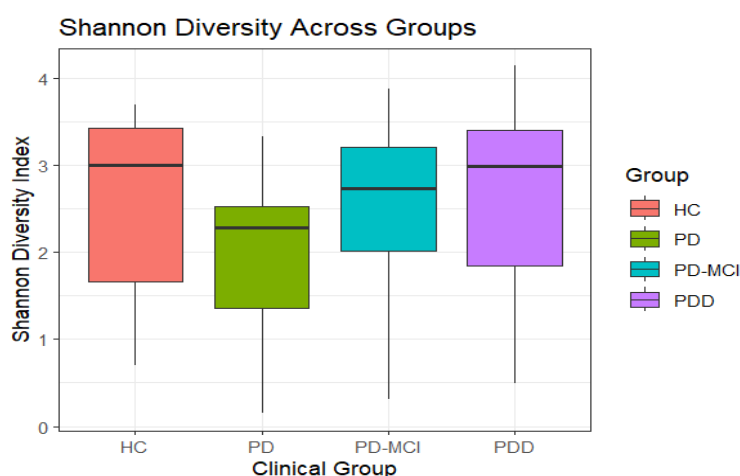


Figure 13. Alpha diversity analysis using Shannon diversity index across clinical groups.

Alpha diversity analysis is widely used to assess the microbial richness and evenness of microbes within the datasets. In PD samples reduced diversity indicates the microbial dysbiosis and decrease in stability linked with Parkinson disease. Variation in diversity patterns observed in both PD-MCI and PDD groups may denoted that progressive restructuring of microbes during cognitive decline and disease advancements. The alteration in diversity of microbes is commonly linked to altered metabolism, inflammatory imbalance and gut barrier disruption in neurodegenerative disorders.

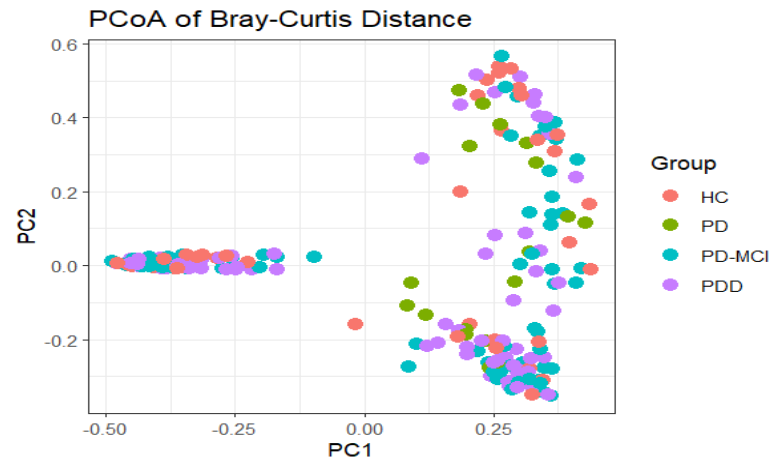


Figure 14. Beta diversity analysis using Bray-Curti's distance-based Principal Coordinate Analysis (PCoA) across clinical groups.

Beta diversity using Bray-Curti's distance and Principal Coordinate Analysis highlighted clear clustering patterns among HC, PD, PD-MCI, and PDD groups. Samples from different groups showed distinct compositional clustering along principal coordinate axes, denoting changes in community structure of microbes in different disease conditions. Shared microbes within the datasets were observed by partial overlaps between several groups.

Beta diversity analysis measures difference in microbes between samples and clinical groups. The clustering among HC, PD, PD-MCI and PDD groups denotes gradual transformation of microbes during disease advancements. Partial overlap in clusters suggest internal microbial variability and shared microbial species across groups. Collectively beta diversity supports the oral-gut-brain axis in Parkinson disease.

6.7 Machine Learning-Based Classification Using XGBoost

```
After SMOTE class balance:
1    188
0    188
Name: count, dtype: int64

Running GridSearchCV...

Best Parameters:
{'colsample_bytree': 0.8, 'learning_rate': 0.02, 'max_depth': 8, 'n_estimators': 800, 'subsample': 0.8}

Accuracy:
0.9292035398230089

Confusion Matrix:
[[53  4]
 [ 4 52]]

Cross-validation accuracy:
[0.71052632 0.90666667 0.92      0.96      0.85333333]
Mean CV accuracy:
0.8701052631578946
```

Figure 15. XGBoost model performance metrics and optimized classification results for Parkinson's disease prediction.

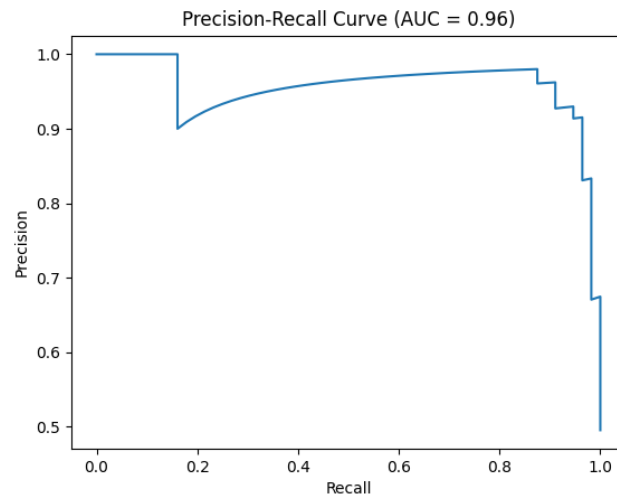


Figure 16. Precision-Recall (PR) curve demonstrating predictive precision and recall performance of the XGBoost classification model.

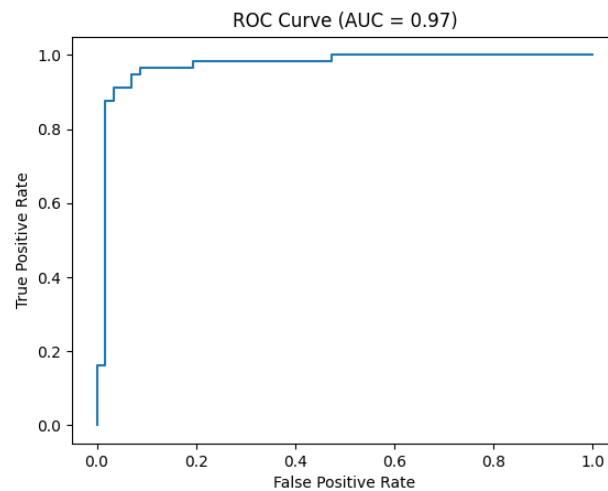


Figure 17. Receiver Operating Characteristic (ROC) curve showing classification performance of the XGBoost model with high AUC value.

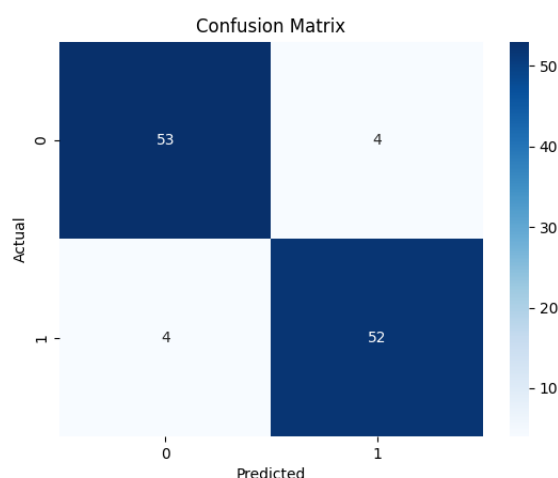


Figure 18. Confusion matrix illustrating accurate classification of Parkinson's disease and healthy control samples using the XGBoost model.

Table 1. Performance evaluation metrics of the XGBoost machine learning model for Parkinson's disease classification

Metric	Value
Accuracy	93%
ROC-AUC	0.97
PR-AUC	0.96
Cross-validation Accuracy	0.87
True Positives	52
True Negatives	53
False Positives	4
False Negatives	4

Machine learning based classification by XGBoost algorithm revealed strong predictive performance for classifying Parkinson disease samples from healthy controls using microbiome related features and OES. After class balancing by SMOTE and GridSearchCV optimization, the model achieved overall accuracy of nearly 93%, with ROC-AUC score of 0.97 and Precision Recall AUC of 0.96. Cross validation revealed a mean accuracy of 0.87, demonstrating robustness of model and generalization capability.

Confusion matrix showed balanced classification performance with 53 True Negatives, 52 True Positives, 4 False Positives, 4 False Negatives. The Precision Recall curve and ROC curve proved high predictive precision and performance of XGBoost model for Parkinson disease classification.

The XGBoost algorithm exhibited high predictive performance for Parkinson disease using microbial abundance and OES as predictive biomarkers. The high ROC-AUC and PR-AUC values suggest strong sensitivity and specificity and classification efficiency. Balanced confusion matrix outcomes support generalizability of model in distinguishing disease vs control samples accurately.

For handling class imbalance SMOTE is utilized to reduce bias in prediction to enhance stability of model and performance. To improve the model prediction accuracy and robustness GridSearchCV optimization was performed for determining the optimal hyperparameters. The XGBoost model achieved a strong performance denotes integration of microbiome signatures with machine learning approaches provide excellent framework for non-invasive early disease diagnosis and monitoring development of disease.

The se denoting growing importance of AI driven microbiome analysis in neurological diseases.

6.8 Machine Learning-Based Classification Using Random Forest

```
After SMOTE class balance:
1    188
0    188
Name: count, dtype: int64

Running GridSearchCV...

Best Parameters:
{'max_depth': 10, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 200}

Accuracy:
0.9026548672566371

Confusion Matrix:
[[53  4]
 [ 7 49]]

Cross-validation accuracy:
[0.75      0.89333333 0.93333333 0.96      0.86666667]
Mean CV accuracy:
0.8806666666666667
PR AUC: 0.9300763454359333
```

Figure 19. Random Forest model optimization and classification performance metrics for Parkinson's disease prediction.

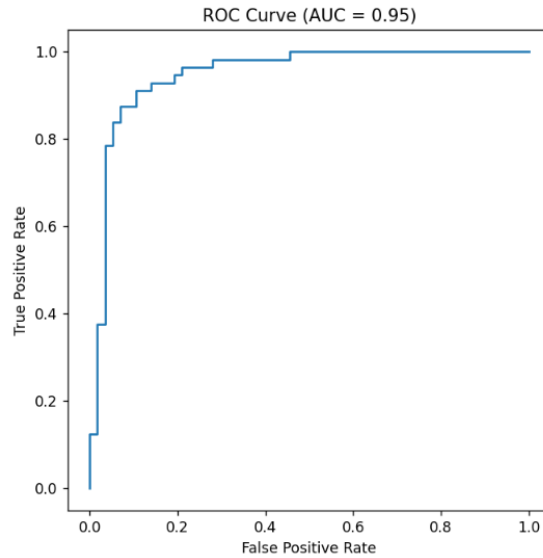


Figure 20. Receiver Operating Characteristic (ROC) curve showing classification performance of the Random Forest model.

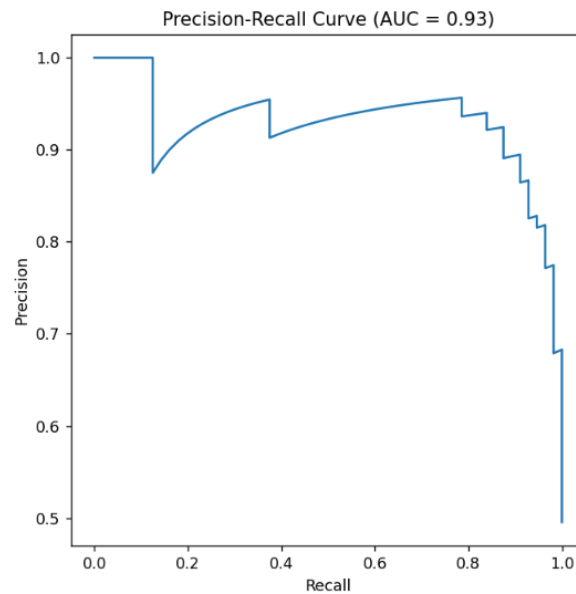


Figure 21. Precision-Recall (PR) curve illustrating predictive precision and recall performance of the Random Forest model.

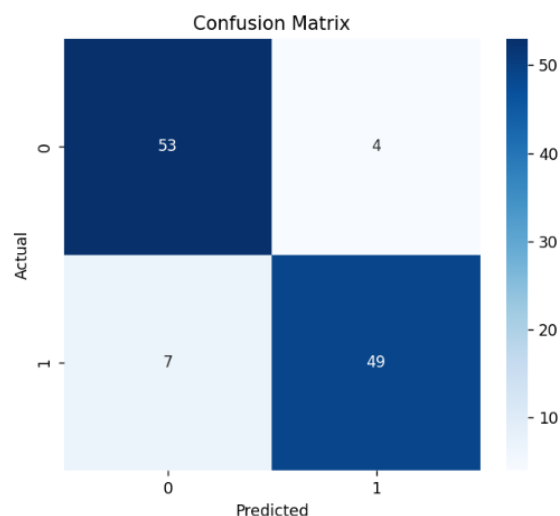


Figure 22. Confusion matrix demonstrating classification accuracy of the Random Forest model

Table 2. Performance evaluation metrics of Random Forest machine learning model for Parkinson’s disease classification

Metric	Value
Accuracy	90%
ROC-AUC	0.95
PR-AUC	0.93
Cross-validation Accuracy	0.88
True Positives	49
True Negatives	53
False Positives	4
False Negatives	7

Random Forest classification model revealed strong predictive performance for classifying Parkinson disease samples from healthy controls using microbiome derived features and OES. After class balancing by SMOTE and GridSearchCV optimization, the model achieved overall accuracy of nearly 90%, with ROC-AUC score of 0.95 and Precision Recall AUC of 0.93. Cross validation revealed a mean accuracy of 0.88, demonstrating robustness of model and generalization capability.

Confusion matrix showed balanced classification performance with 53 True Negatives, 49 True Positives, 4 False Positives, 7 False Negatives. The Precision Recall curve and ROC curve proved high predictive precision and performance Random Forest model for Parkinson disease classification.

The Random Forest model exhibited high predictive performance for Parkinson disease using microbial abundance and OES as predictive biomarkers. The high ROC-AUC and PR-AUC values suggest strong sensitivity and specificity and classification efficiency.

Balanced confusion matrix outcomes support generalizability of model but slightly high number of false negatives observed compared to XGBoost model.

For handling class imbalance SMOTE is utilized to reduce bias in prediction to enhance stability of model and performance. To improve the model prediction accuracy and robustness GridSearchCV optimization was performed for determining the optimal hyperparameters.

The Random Forest model achieved a strong predictive performance, the XGBoost model obtained superior classification accuracy and performance compared to Random Forest model. Both ML models denoted the AI approaches for early disease diagnosis and disease monitoring in Parkinson's.

6.8 Identification of Key Microbial Biomarkers

XGBoost identified several microbes strongly linked with Parkinson disease. Among all top ranked biomarkers *Streptomyces albidoflavus* displayed higher feature importance score on followed by *Akkermansia* sp. BCRC 18949, *Streptococcus* sp. oral taxon 431, *Porphyromonas endodontalis*, *Prevotella histicola*, and multiple *Streptococcus*-associated taxa. Further significant biomarkers obtained are *Actinomyces* species, *Schaalia* species, and oral-associated *Prevotella* species. The identified biomarkers denote strong differentiation capability between Parkinson disease and healthy controls samples.

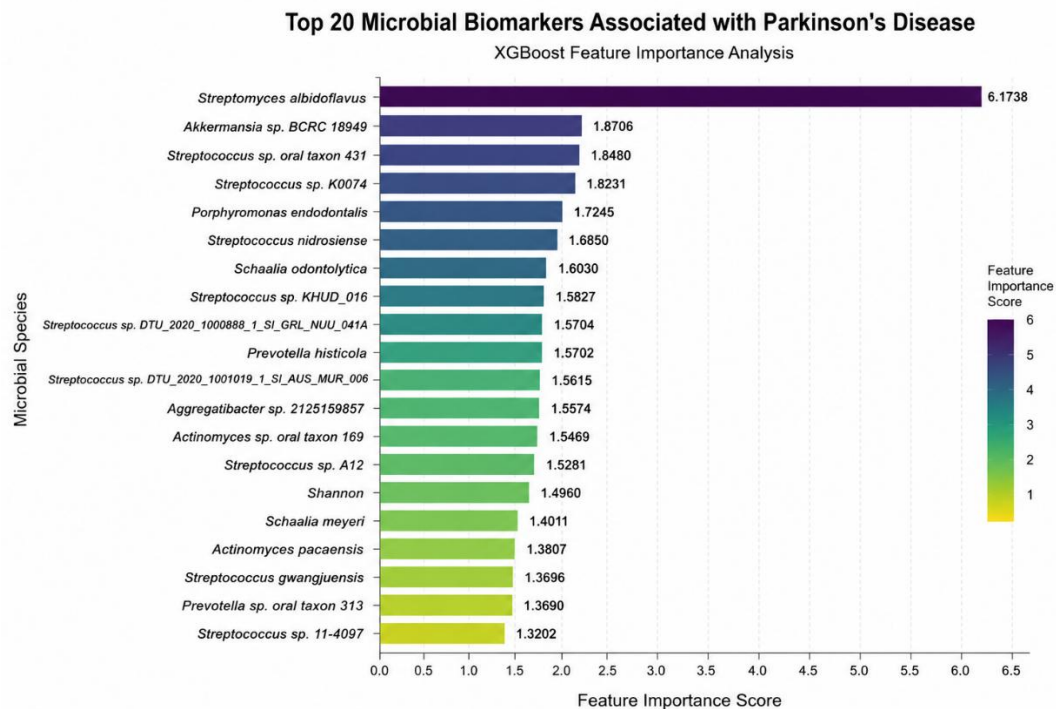


Figure 23. XGBoost feature importance analysis identifying top microbial biomarkers associated with Parkinson's disease.

The identified microbial biomarkers showed strong association with Parkinson's disease and denoted importance in oral-gut microbial dysbiosis. *Streptococcus*, *Porphyromonas*, and *Actinomyces* species were the dominant taxa and are linked with biofilm formation, immune modulation and inflammation. The dominance of oral-associated microbes supports oral-to-gut microbial translocation and leads to disease development. *Akkermansia* sp. and *Prevotella* species biomarkers suggest variation in gut barrier integrity and in oral-gut-brain axis there is a metabolic imbalance.

6.9 Virulence Factor Analysis

Virulence category distribution showed adhesion-linked virulence factors are more dominant among Parkinson disease-associated microbes. In microbial biomarkers categories such as immune modulation and toxin-related categories are comparatively higher. Along with that virulence-linked mechanisms such as secretion systems, stress response, redox regulation, and host signaling alterations were present in lower numbers.

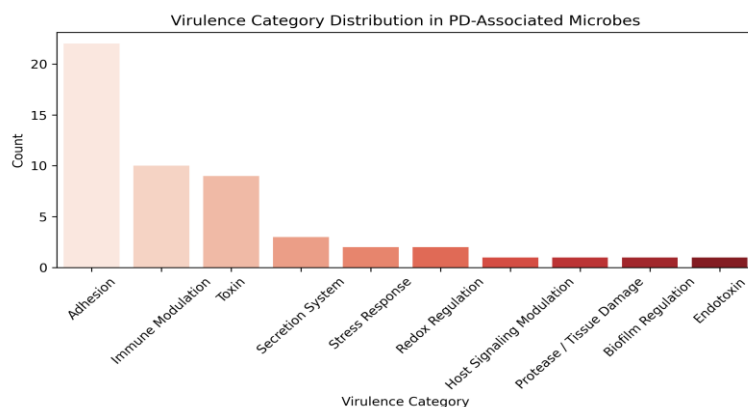


Figure 24. Distribution of virulence factor categories identified among Parkinson's disease-associated microbial biomarkers.

The dominance of adhesion related virulence factors denotes increased microbial colonization and biofilm producing capability in Parkinson disease associated microbes. The inflammatory and host damaging microbial activities within oral-gut-brain axis is due to enrichment of mechanisms linked with immune alteration and toxin. During disease development this virulence linked factors leads to the chronic inflammation, microbial persistence and gut barrier dysregulation

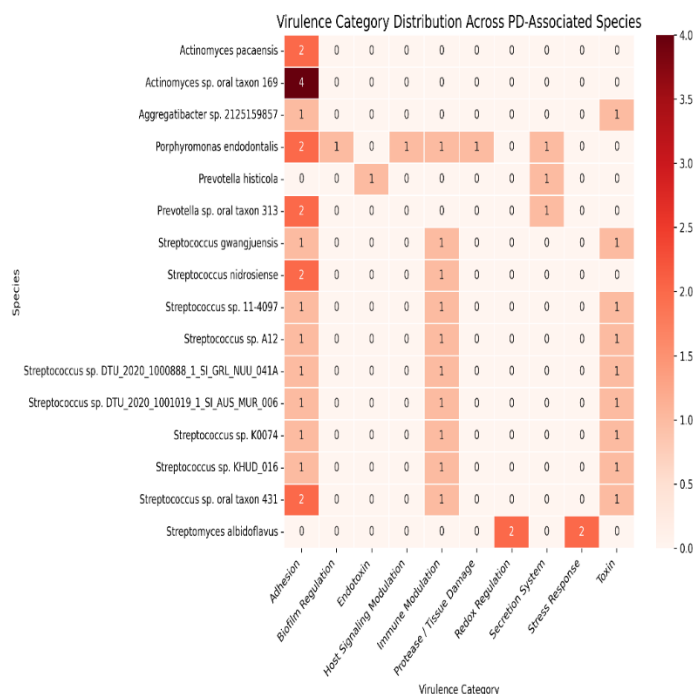


Figure 25. Heatmap showing species-wise distribution of virulence factor categories across PD-associated microbial taxa.

Virulence heatmap at species level highlighted distinct virulence patterns across Parkinson's disease associated biomarkers. Adhesion, immune modulation, toxin-linked activity, and stress response pathways were shown enrichment by microbial species such as *Porphyromonas endodontalis*, *Streptococcus* species, *Actinomyces* species, and *Prevotella* species. Some taxa additionally shown linked with secretion system and host signaling related virulence factors. The heatmap exhibits species specific variation in virulence factor patterns across found microbial biomarkers. The presence of stress response pathways and toxin linked activities further confirms pathogens involvement in Parkinson disease development by contributing oral-gut-microbial dysbiosis and neuroinflammation.

6.12 Functional Pathway Mapping of PD-Associated Microbial Biomarkers

Metabolic Pathways and Functional Roles of PD-Associated Microbial Biomarkers

Biomarker	MetaCyc Pathway	Functional Meaning	Pathway Category
<i>Streptomyces albidoflavus</i>	Secondary metabolite biosynthesis	Antibiotics / host interaction	Secondary Metabolism
<i>Streptomyces albidoflavus</i>	Redox metabolism	Oxidative stress	Energy Metabolism
<i>Akkermansia</i> sp. BCRC 18949	Mucin degradation	Gut barrier modulation	Carbohydrate Metabolism
<i>Akkermansia</i> sp. BCRC 18949	SCFA production	Gut brain signaling	Short-chain Fatty Acid (SCFA)
<i>Streptococcus</i> sp. oral taxon 431	Lactic acid fermentation	Acid production	Carbohydrate Metabolism
<i>Streptococcus</i> sp. oral taxon 431	Biofilm formation	Oral colonization	Cellular Processes
<i>Streptococcus</i> sp. K0074	Lactic acid fermentation	Acid production	Carbohydrate Metabolism
<i>Streptococcus</i> sp. K0074	Adhesion protein secretion	Biofilm	Cellular Processes
<i>Porphyromonas endodontalis</i>	LPS biosynthesis	Inflammation	Cellular Processes
<i>Porphyromonas endodontalis</i>	Heme acquisition pathway	Survival in host	Transport & Binding Proteins
<i>Porphyromonas endodontalis</i>	Proteolysis pathway	Tissue damage	Amino Acid Metabolism
<i>Streptococcus nidrosiense</i>	Lactic acid fermentation	Acid production	Carbohydrate Metabolism
<i>Schaalia odontolytica</i>	Amino acid metabolism	Oral biofilm support	Amino Acid Metabolism
<i>Streptococcus</i> sp. KHUD_016	Lactic acid fermentation	Acid production	Carbohydrate Metabolism
<i>Streptococcus</i> sp. DTU_2020_1000888_1_SI_GRL_NUU_041A	Lactic acid fermentation	Acid production	Carbohydrate Metabolism
<i>Prevotella histicola</i>	Lactic acid fermentation	Acid production	Carbohydrate Metabolism
<i>Prevotella histicola</i>	Carbohydrate fermentation	SCFA production	Carbohydrate Metabolism
<i>Streptococcus</i> sp. DTU_2020_1001019_1_SI_AUS_MUR_006	Type VI secretion system	Microbial competition	Cellular Processes
<i>Aggregatibacter</i> sp. 2125159857	Lactic acid fermentation	Acid production	Carbohydrate Metabolism
<i>Aggregatibacter</i> sp. 2125159857	Biofilm formation	Colonization	Cellular Processes
<i>Actinomyces</i> sp. oral taxon 169	Iron acquisition system	Survival	Transport & Binding Proteins
<i>Actinomyces</i> sp. oral taxon 169	Biofilm formation	Early colonizer	Cellular Processes
<i>Actinomyces</i> sp. oral taxon 169	Amino acid metabolism	Growth support	Amino Acid Metabolism
<i>Streptococcus</i> sp. A12	Lactic acid fermentation	Acid production	Carbohydrate Metabolism
<i>Schaalia meyeri</i>	Amino acid metabolism	Oral ecosystem	Amino Acid Metabolism
<i>Actinomyces pacaensis</i>	Biofilm formation	Colonization	Cellular Processes
<i>Streptococcus gwanguensis</i>	Lactic acid fermentation	Acid production	Carbohydrate Metabolism
<i>Prevotella</i> sp. oral taxon 313	Biofilm formation	Persistence	Cellular Processes
<i>Prevotella</i> sp. oral taxon 313	Carbohydrate metabolism	Fermentation	Carbohydrate Metabolism
<i>Streptococcus</i> sp. 11-4097	Lactic acid fermentation	Acid production	Carbohydrate Metabolism

Figure 26. Functional pathway mapping and MetaCyc-based metabolic annotation of Parkinson's disease-associated microbial biomarkers.

Functional pathway mapping was performed by MetaCyc tool identified some metabolic and host interaction pathways linked with Parkinson's disease. Key pathways obtained are lactic acid fermentation, biofilm formation, amino acid metabolism, short-chain fatty acid (SCFA) production and lipopolysaccharide synthesis. Several microbial species like *Porphyromonas endodontalis*, *Akkermansia* sp. BCRC 18949, *Streptococcus* species, and *Actinomyces* species were associated with oral colonization, tissue damage,

oxidative stress, inflammation and gut barrier alterations. These indicate that oral microbial persistence and oral to gut microbial translocation in Parkinson disease. Further pathways linked with amino acid metabolism and microbial competition highlight host microbial interaction linked to neuroinflammation and metabolic dysregulation. Overall, the findings support the functional importance of PD associated microbial biomarkers and their contribution to Parkinson's disease progression.

6.13 Co-Integrated Microbial Mechanistic Network Analysis

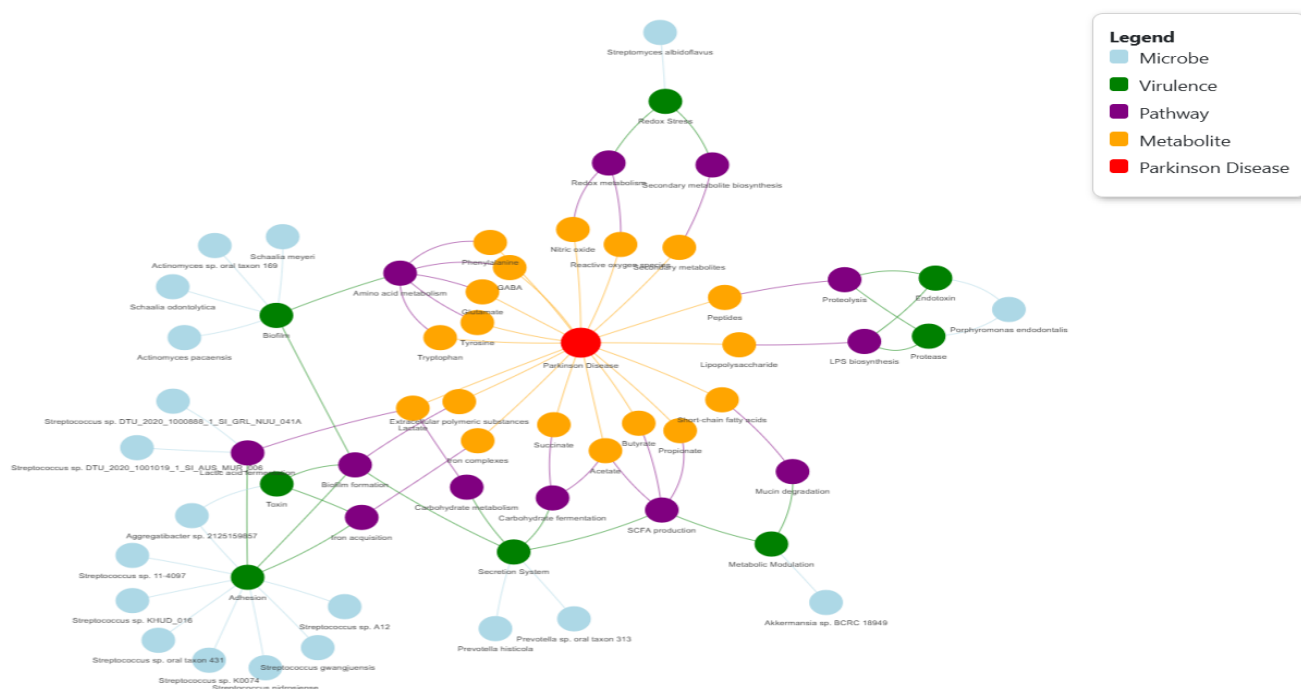


Figure 27. Co-integrated microbial mechanistic network linking microbial biomarkers, virulence factors, metabolic pathways, metabolites, and Parkinson's disease associations.

The Co-integrated microbial mechanistic network analysis demonstrated complex interactions between microbial biomarkers, virulence factors, inferred metabolites, metabolic pathways and Parkinson's disease linked mechanisms. Some oral associated microbes such as *Streptococcus*, *Porphyromonas*, *Actinomyces*, *Prevotella*, and *Akkermansia* species were connected together pathways involved in mucin degradation, and lipopolysaccharide (LPS) biosynthesis, SCFA production, amino acid metabolism and biofilm formation. In additionally virulence mechanisms were also demonstrated in the network such as toxin, adhesion, secretion system and immune modulations. Parkinson disease appeared as central interacting node connecting microbial dysbiosis, metabolic pathways, virulence activities. Associations within microbial pathways and metabolites suggesting potential impact of neuro immune signaling and host metabolic homeostasis. Collectively the network analysis revealed how oral and gut microbial dysbiosis all together contribute to neuroinflammation and Parkinson's disease progression through coordinated mechanistic interactions.

CONCLUSION

In this study, the oral-gut-brain axis in Parkinson's disease was successfully investigated by using an integrated multi-omics and machine learning framework based on paired metagenomic shotgun datasets from oral and gut samples. The analysis revealed significant microbial dysbiosis linked with the progression of Parkinson's disease and highlighted the oral to gut microbial translocation role in neurodegenerative processes.

Several microbial species from oral cavity including *Streptomyces albidoflavus*, *Porphyromonas endodontalis*, *Akkermansia species*, *Streptococcus species*, and *Actinomyces species*, were identified as significant microbial biomarkers associated with Parkinson's disease progression and cognitive impairments. Quality analysis and taxonomic profiling confirmed the reliability of metagenomic datasets and revealed dominant microbial species linked with inflammation and metabolic disorder.

Diversity analysis revealed progressive changes in composition of microbes in healthy controls as well as in groups with Parkinson's disease, PD-MCI, and PDD. The Oral Enrichment Score (OES) demonstrated that during the disease advancement there was increase in microbial abundance from Oral cavity, supporting the hypothesis of microbial translocation from oral cavity to gut in Parkinson's disease.

Metabolic pathway analysis and virulence analysis revealed group of metabolic pathways linked with lipopolysaccharide biosynthesis, amino acid metabolism, biofilm formation, adhesion, immune modulation, secretion system, oxidative stress, toxin-related mechanisms. These findings suggest the microbial dysbiosis may contribute to neuroinflammation, intestinal barrier integrity, and metabolic imbalance in host via coordinated microbial interactions within the oral-gut-brain axis.

The machine learning based biomarker identification showed excellent predictive performance in Parkinson's disease classification. Among the models examined, XGBoost achieved superior classification accuracy with higher ROC-AUC and PR-AUC values, highlighting the significance of incorporating microbial abundance profiles, diversity, functional pathways, virulence patterns, and OES for disease prediction.

The identified microbial biomarkers demonstrated high ability to distinguish samples between healthy controls and Parkinson's disease patients, underscoring their utility as non-invasive diagnostic biomarkers. Overall, this study gives evidence for mechanistic role of oral-gut-brain axis in progression of Parkinson's disease.

The integration of metagenomics, functional pathway analysis, virulence analysis, network analysis, and AI-based biomarker discover gives a robust framework for understanding role of oral microbes in neurodegeneration. These findings could lead to the development of microbiome based diagnostic strategies, precision medicine approaches, and therapeutic interventions targeting microbial dysbiosis in Parkinson's disease.

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